Dr. Rob van Eijk
Future of Privacy Forum

On Privacy & Explainable AI/ML

22 June 2022
Data Science Technical Track

Universiteit Leiden
Leiden Centre of Data Science

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Two topics

- A privacy framing for AI/ML
- Explainability for AI/ML

A definition for privacy

„Privacy is the claim of individuals, groups, or institutions to determine for themselves when, how and to what extent information about them is communicated.”

[Westin, 1967, p. 7]

Questions:
- What does privacy mean nowadays to you?
- To what extent is data intertwined with your work?

Four Building Blocks [1]

The right to privacy in Europe is based upon four building blocks.

Building block 1:

- Article 16 of the **Treaty on the Functioning of the European Union (TFEU)** which provides the legal basis for the adoption of Union legal instruments relating to the protection of personal data.

Four Building Blocks [2]

Building block 2 and 3:

- Article 7 and Article 8 of the **Charter of Fundamental Rights of the European Union (CFREU)**, the charter has a similar legal value as the TFEU.

We will look at both articles

Four Building Blocks [3]

- **Article 7**: „*Everyone has the right to respect for his or her private and family life, home and communications.*"

Four Building Blocks [4]

- **Article 8** sees to the protection of personal data:

„(1) Everyone has the right to the protection of personal data concerning him or her,

(2) such data must be processed fairly for specified purposes and on the basis of the consent of the person concerned or some other legitimate basis laid down by law. Everyone has the right of access to data which has been collected concerning him or her, and the right to have it rectified,

(3) compliance with these rules shall be subject to control by an independent authority."

Four Building Blocks [5]

Building block 4:

- **General Data Protection Regulation (EU) 2016/679 (GDPR)** repealing General Data Protection Directive 95/46/EC (GDPD).
- The application and interpretation of the legal norms in the GDPR **conform to the CFREU**.

Eight OECD principles (1980, 2013)

- Collection limitation
- Data quality
- Purpose specification
- Use limitation
- Security safeguards
- Openness
- Individual participation, and
- Accountability

The GDPR aligns with the OECD principles

- Lawfulness, fairness, and transparency (Article 5(1a), Art. 6 GDPR)
- Purpose limitation (Article 5(1b) GDPR)
- Data minimization (Article 5(1c) GDPR)
- Accuracy (Article 5(1d) GDPR)
- Storage limitation (Article 5(1e) GDPR)
- Integrity and confidentiality (Article 5(1f) GDPR), and
- Accountability (Article 5(2) GDPR)

Privacy and other fundamental rights

- Freedom of expression, religion, and assembly/association
- Non-discrimination
- Presumption of innocence/right of defense
- Freedom of speech, freedom of thought, freedom of movement, and
- Right to liberty

Data protection in court

- EU law supersedes national law.
- A judge will apply a legal norm against the background of the aforementioned principles and fundamental rights.

SyRI legislation in violation of privacy

- ECLI:NL:RBDHA:2020:865
- Cf. Article 8, paragraph 2 of the European Convention on Human Rights
- “The Court is of the opinion that the SyRI legislation does not provide sufficient safeguards to protect the right to respect for private life in relation to the risk indicators and the risk model that can be implemented in a specific SyRI project.”

Personal Data [1]

Article 4(1) GDPR:

Any information relating to an identified or identifiable natural person ('data subject');

But… what is an identifiable natural person?

Personal Data [2]

Article 4(1) GDPR:

An identifiable natural person is one who can be identified, **directly or indirectly**,

in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or

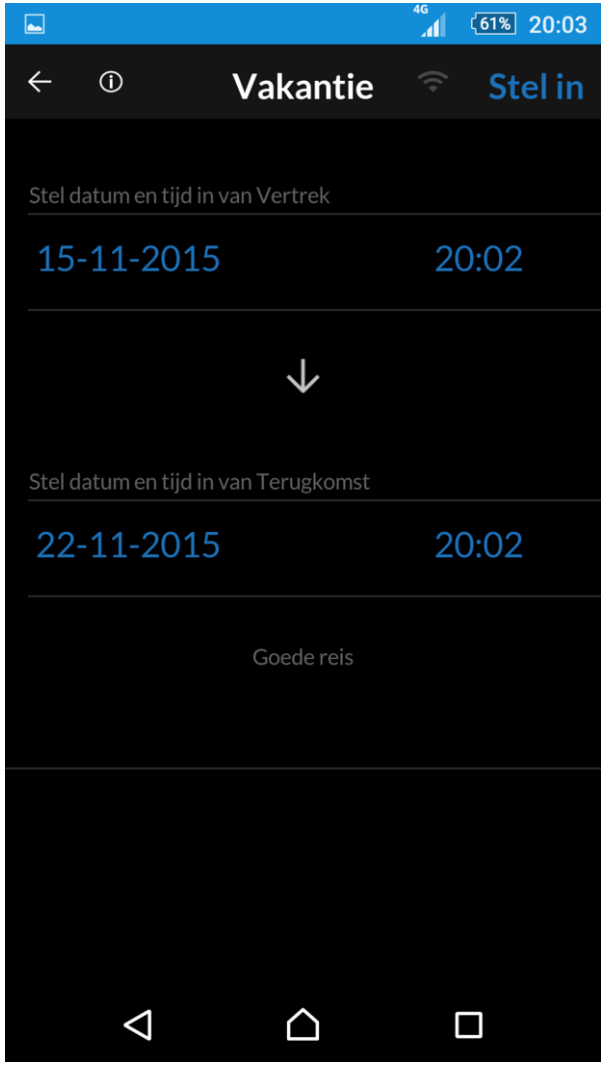
to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person.

Personal Data [3]

Recital 26 GDPR: To determine whether a natural person is identifiable, account should be taken of **all the means reasonably likely to be used**, such as singling out, either by the controller or by another person to identify the natural person directly or indirectly.

Note: in the online context personal data is often data about your behavior, i.e., (1) ***what*** you use and (2) ***how*** you use it.

Question: personal data?



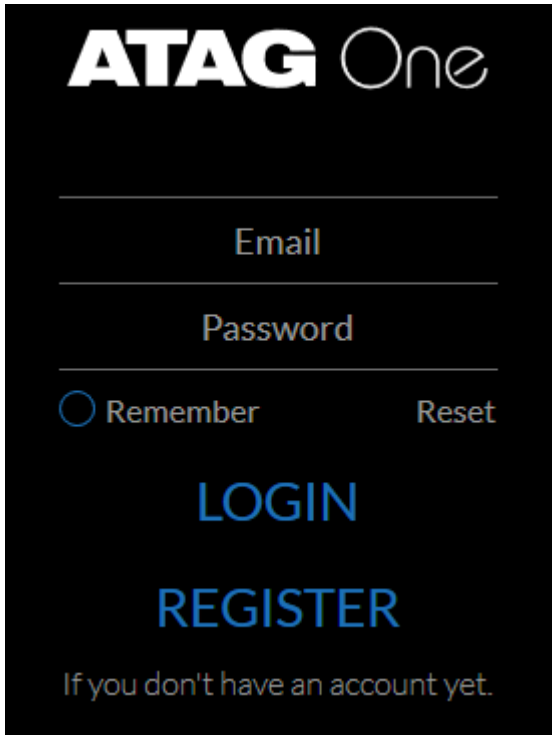
Meta data about my behavior

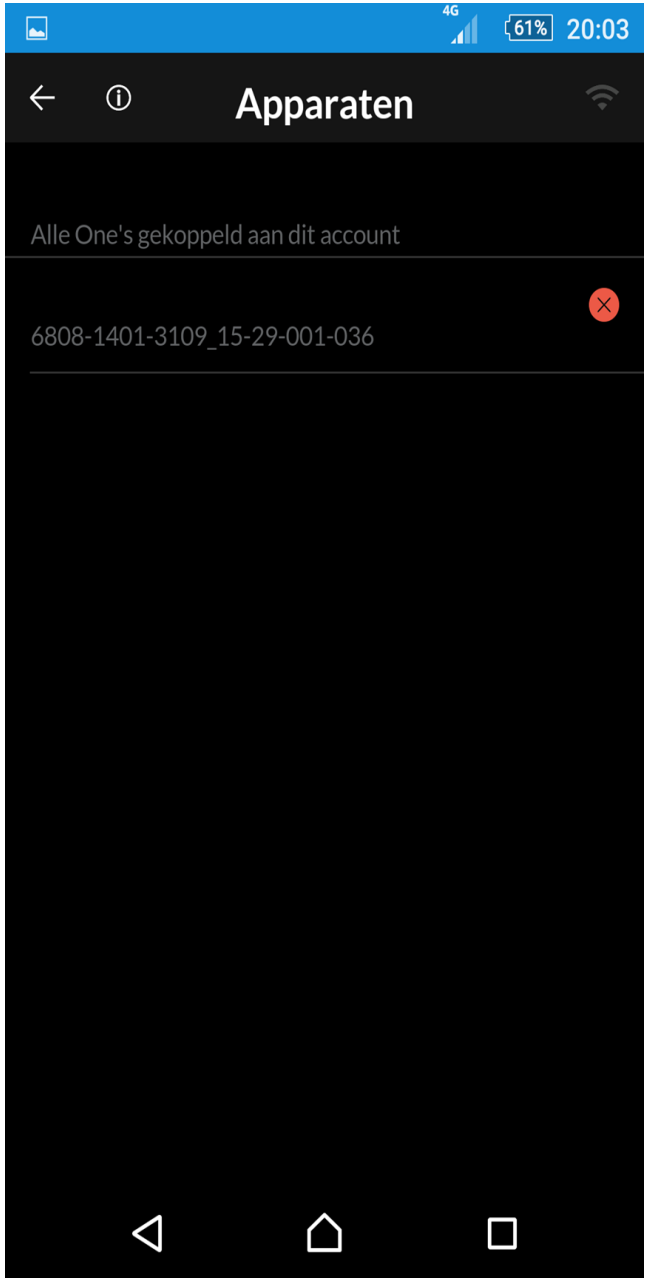
Device data:

- Type
- Heating capacity

Operational data:

- Date, time
- Burning hours
- Burning status
- Room temperature





Privacy: a risk based approach [1]
(Recital 75 GDPR)

- The risk to the **rights and freedoms of natural persons**,
- of varying likelihood and severity, may result from personal data processing which could lead to
  - physical,
  - material or
  - non-material damage, in particular:

Privacy: a risk based approach [2]
(Recital 75 GDPR)

- where the processing may give rise to
  - discrimination,
  - identity theft or fraud,
  - financial loss,
  - damage to the reputation,
  - loss of confidentiality of personal data protected by professional secrecy,
  - unauthorized reversal of pseudonymization, or
  - any other significant economic or social disadvantage;

Privacy: a risk based approach [3]
(Recital 75 GDPR)

- where data subjects might be deprived of their rights and freedoms or prevented from exercising control over their personal data;
- where personal data are processed which reveal **racial or ethnic origin, political opinions, religion or philosophical beliefs, trade union membership**,
- and the processing of **genetic data, data concerning health or data concerning sex life or criminal convictions and offences or related security measures**;

Privacy: a risk based approach [4]
(Recital 75 GDPR)

- where personal aspects are evaluated, in particular analyzing or predicting aspects concerning **performance at work, economic situation, health, personal preferences or interests, reliability or behavior, location or movements**, in order to create or use personal **profiles**;
- where personal data of vulnerable natural persons, in particular of **children**, are processed; or
- where processing involves a **large amount** of personal data and affects a **large number** of data subjects.

Pseudonymization [1]
(Article 4(5) GDPR)

- the processing of personal data in such a manner that the personal data can no longer be attributed to a specific data subject **without the use of additional information**,
- provided that such additional information is kept separately and is subject to technical and organizational measures **to ensure that the personal data are not attributed to an identified or identifiable natural person**.

Pseudonymization [2]
(Article 4(5) GDPR)

Personal data which have undergone pseudonymization, which could be attributed to a natural person by the use of additional information **should be considered to be information on an identifiable natural person**.

Anonymization [1]

- The principles of data protection should therefore **not apply to anonymous information**.
- Namely information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous **in such a manner that the data subject is not or no longer identifiable**.
- This Regulation does not therefore concern the processing of such anonymous information, **including for statistical or research purposes**.

Anonymization [2]

WP29 Opinion 5/2014 on Anonymization Techniques:

- Anonymization constitutes **a further processing of personal data**; as such, it must satisfy the requirement of compatibility by having regard to the legal grounds and circumstances of the further processing.

Anonymization [3]

WP29 Opinion 5/2014 on Anonymization Techniques :

The opinion elaborates on the robustness of each technique based on three criteria:

- is it still possible to **link records relating to an individual**, and
- is it still possible to **single out** an individual,
- can information be **inferred concerning an individual**?

Two topics

- A privacy framing for AI/ML
- Explainability for AI/ML

Privacy framing for AI issues – “FAT” (1)

- The discussion of AI privacy implications has largely been framed around the concepts of **Fairness**, **Accountability**, and **Transparency** (frequently shortened to “FAT”).
- Taken together, these concepts represent some of the unique aspects of machine learning models that raise the most concern.

[Future of Privacy Forum (2018). *The Expert’s Guide To Artificial Intelligence and Machine Learning,* p. 22]

Privacy framing for AI issues – “FAT” (2)

- **Fairness**: Ensure that algorithmic decisions do not create discriminatory or unjust impacts when comparing across different demographics or affected communities and individuals.
- **Accountability**: Algorithmically informed decisions have the potential for significant social impact, and must be designed and implemented in publicly accountable ways, such as an obligation to report, explain and justify specific decisions as well as mitigate negative impacts and potential harms.
- **Transparency**: ensuring that the personal data collected, the model’s function and process, and the role of the model’s output in final decision-making are explainable to the user or individual impacted by the decision.

Privacy framing for AI issues – “FAT” (3)

- Fairness is a highly prized human value.
- **Inequality** and **unfairness** have complex causes, but bias in the decisions that organizations make about individuals is often a key aspect.
- **Transparency** is key in building and maintaining trust.
- Requirements for transparency:
  - Analytical techniques to expose **bias**,
  - **Ethical** and **legal considersations** that inform best responses.

[Centre for data ethics and innovation (2020). *Review in to bias in algorithmic decision-making*, pp. 3-13, 21-36]

Explainability and transparency (1)

- While not interchangeable, these two concepts are related.
- **Explainability** describes the privacy risk discussions that need to occur when decisions are made in the development and design stages of a ML model.
- When decisions made by the model will affect individuals or members of a group, and are reached based solely on the model output, without human review or intervention, there is a potential for problems.

[Future of Privacy Forum (2018). *The Expert's Guide To Artificial Intelligence and Machine Learning,* p. 17]

Explainability and transparency (2)

- **Transparency** is generally the ability to interpret the functioning of machine learning models, provide explanations for specific algorithmic outputs, and describe or provide frameworks for conducting ethical and legal reviews and audits.
- It might include providing data subjects with process details, informing them when a decision was reached via an automated system, and providing an understanding of the safeguards and rights available to them.

Explainability and transparency (3)

- The reality is that many cutting edge machine learning models include multi-layer neural networks which follow as an internal process under which a particular outcome may not be able to be understood in a mathematically detailed, replicable way even by the data scientists and designers.
- But the multi-layer feature in fact may make the model “better” in terms or relevance or applicability.
- This is recognized as the accuracy vs. interpretability tradeoff (in terms of explainability and transparency).
- It is also sometimes called **the “black box” problem.**

FAT/ML

- Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

Explainability

Ensure that algorithmic decisions as well as any data driving those decisions can be explained to end-users and other stakeholders in non-technical terms.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

FAT/ML

- Principle
  Impact S

Explainability

Ensure that algorithmic decisions as
other stakeholders in non-technical

Explainability

Guiding Questions

- Who are your end-users and stakeholders?
- How much of your system / algorithm can you explain to your users and stakeholders?
- How much of the data sources can you disclose?

Initial Steps to Take

- Have a plan for how decisions will be explained to users and subjects of those decisions. In some cases it may be appropriate to develop an automated explanation for each decision.
- Allow data subjects visibility into the data you store about them and access to a process in order to change it.
- If you are using a machine-learning model:
  - consider whether a directly interpretable or explainable model can be used.
  - describe the training data including how, when, and why it was collected and sampled.
  - describe how and when test data about an individual that is used to make a decision is collected or inferred.
- Disclose the sources of any data used and as much as possible about the specific attributes of the data. Explain how the data was cleaned or otherwise transformed.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

FAT/ML

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Explainability

Fairness

Ensure that algorithmic decisions do not create discriminatory or unjust impacts when comparing across different demographics (e.g. race, sex, etc).

We have left some of the terms above purposefully under-specified to allow these principles to be broadly applicable. Applying these principles well should include understanding them within a specific context. We also suggest that these issues be revisited and discussed throughout the design, implementation, and release phases of development. Two important principles for consideration were purposefully left off of this list as they are well-covered elsewhere: privacy and the impact of human experimentation. We encourage you to incorporate those issues into your overall assessment of algorithmic accountability as well.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

FAT/ML

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Fairness

Guiding Questions

- Are there particular groups which may be advantaged or disadvantaged, in the context in which you are deploying, by the algorithm / system you are building?
- What is the potential damaging effect of uncertainty / errors to different groups?

Initial Steps to Take

- Talk to people who are familiar with the subtle social context in which you are deploying. For example, you should consider whether the following aspects of people's identities will have impacts on their equitable access to and results from your system:
- Race
- Sex
- Gender identity
- Ability status
- Socio-economic status
- Education level
- Religion
- Country of origin
- If you are building an automated decision-making tool, you should deploy a fairness-aware data mining algorithm. (See, e.g., the resources gathered at http://fatml.org).
- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

- Making sense of the black box
  - **LIME**
  - Challenges and pitfalls
  - How do we evaluate explanations?
  - What makes a good explanation?

How do we make sense of the black box? (1)

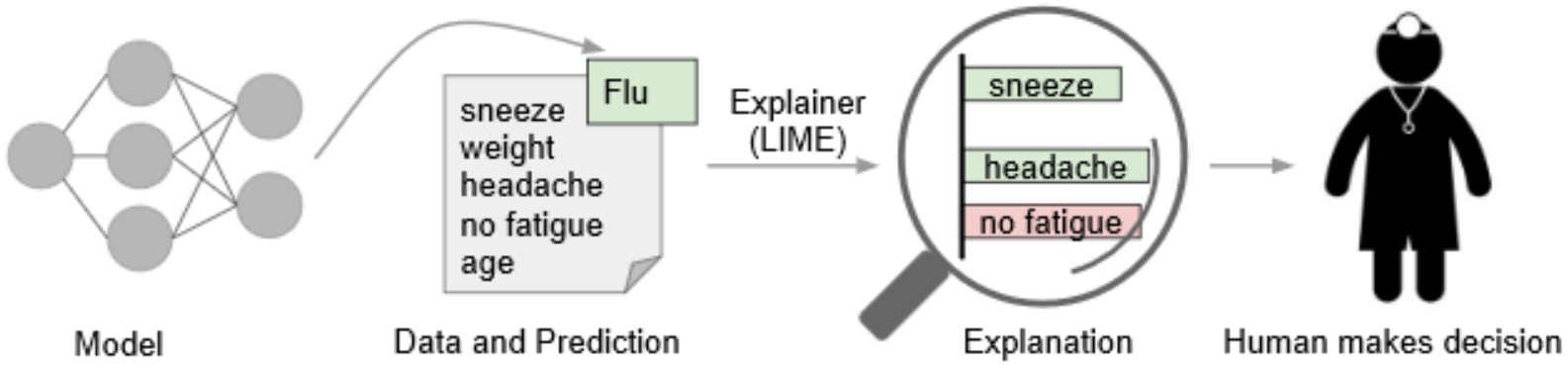
"Why should I trust you?"
Explaining the predictions of any classifier

Video: https://www.youtube.com/watch?v=hUnRCxnydCc en https://edu.nl/3wf9r

edu.nl/3wf9r

How do we make sense of the black box? (2)

- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining,* pp. 1135-1144.
- We want to characterize the relationship between the data and our outcome.

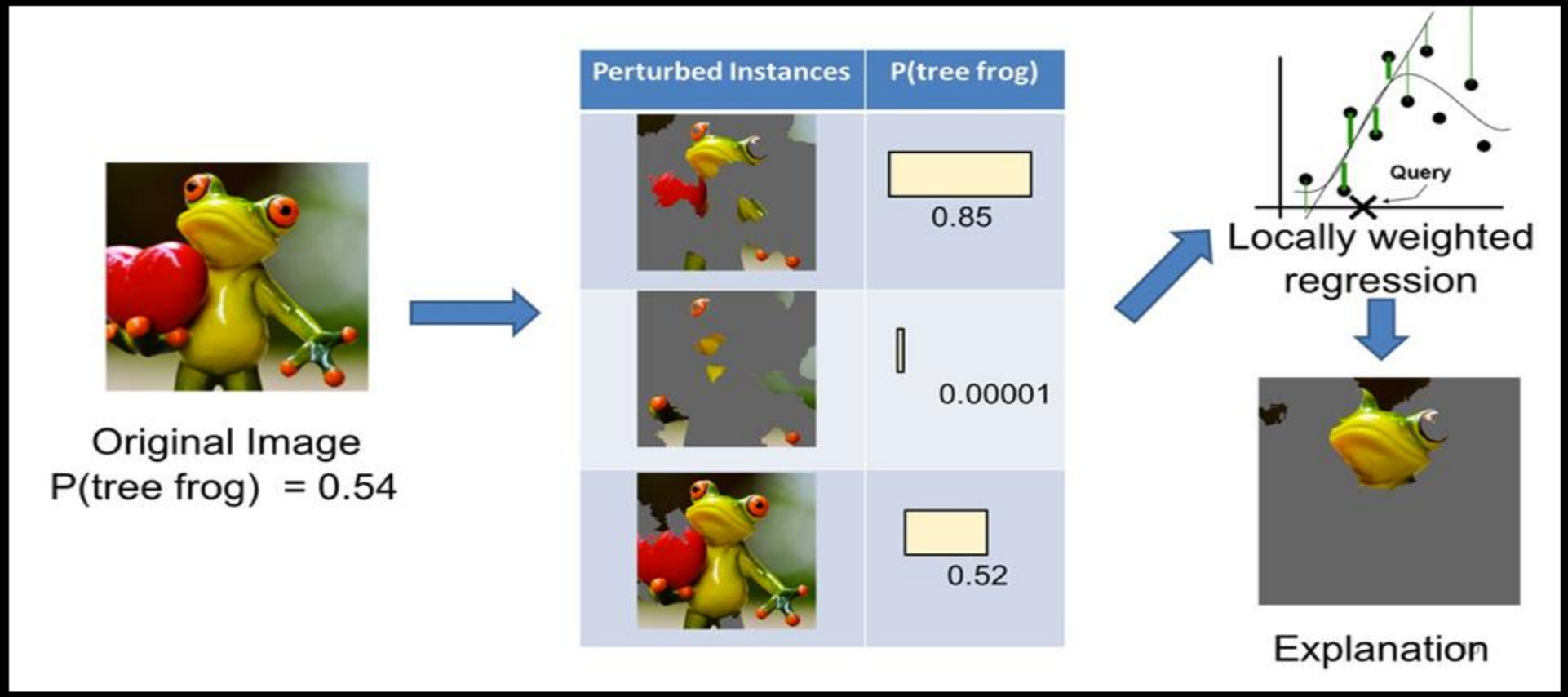


How do we make sense of the black box? (3)

- There are a number of exploratory processes being developed to provide understanding or interpretability of extremely complex machine learning models.
- LIME purports to explain the predictions of any classifier.
- LIME stands for:
  - Local
  - Interpretable
  - Model agnostic
  - Expanations

[Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining,* pp. 1135-1144]

“LIME” (Locally Interpretable Model-agnostic Explanations



Why Should I Trust You? Explaining the Decisions of Any Classifier
Ribeiro et al. 2016

https://towardsdatascience.com/understanding-how-lime-explains-predictions-d404e5d1829c

[Tomsett, R. (2020). Explainable AI (XAI). In *Digital Data Flows Masterclass: Understanding Machine Learning*, Future of Privacy Forum, CPDP 2020 side event]

(a) Husky classified as wolf

(b) Explanation

Why Should I Trust You? Explaining the Decisions of Any Classifier
Ribeiro et al. 2016

Why does the machine think this man isn't smiling? Show a "contrastive" explanation:

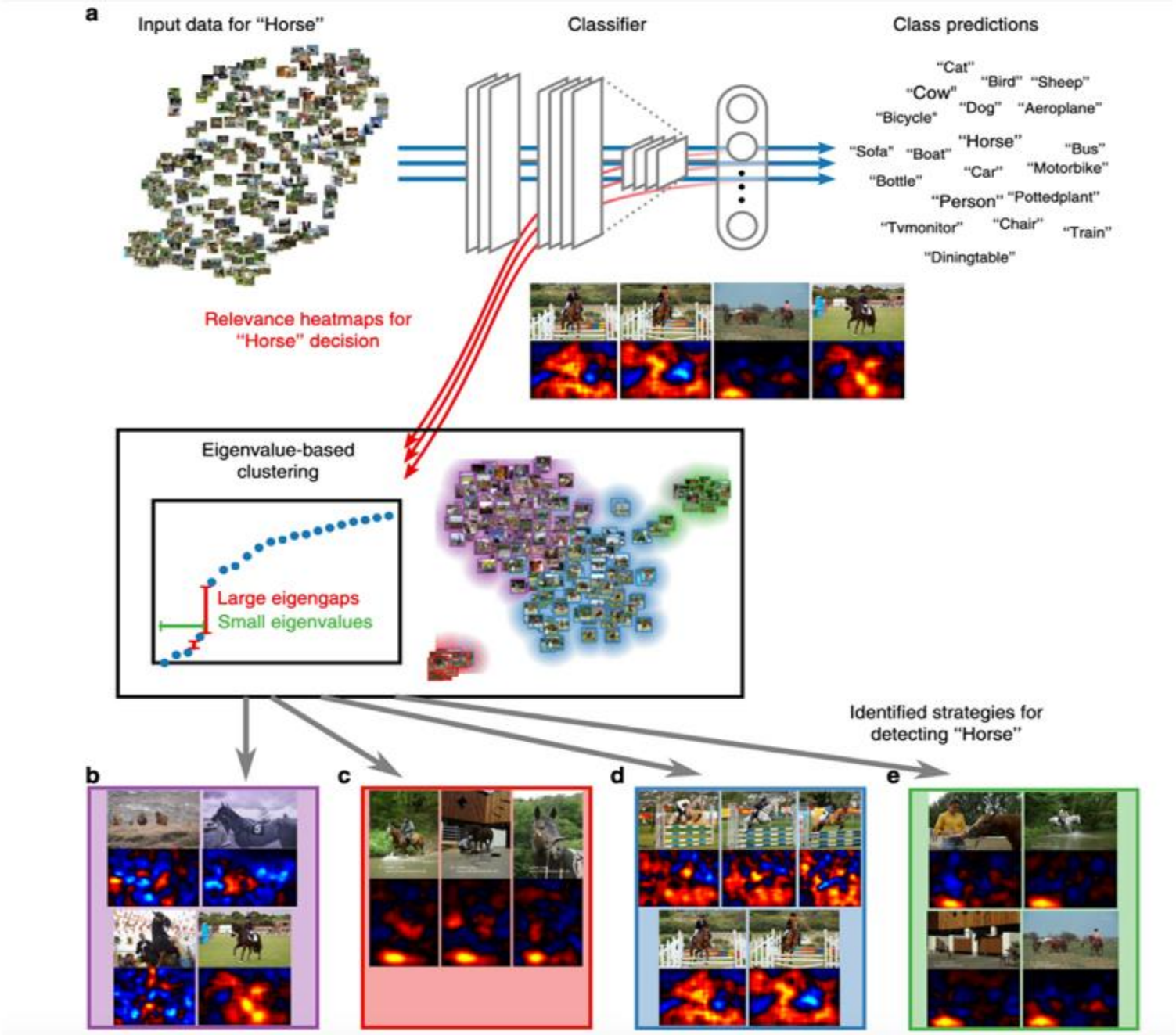
Generating Contrastive Explanations with Monotonic Attribute Functions
Luss et al. (2019)

Original -
Male: 1
Smile: 0
Young: 0

Explanation -
Male: 1
Smile: 1
Young: 0

- Making sense of the black box
  - LIME
  - **Challenges and pitfalls**
  - How do we evaluate explanations?
  - What makes a good explanation?

Spectral Relevance Analysis



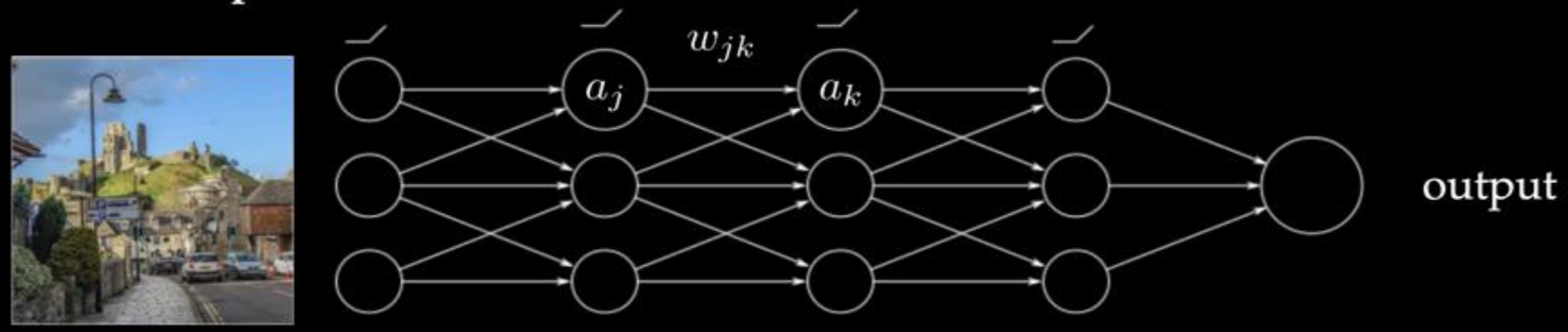
Lapuschkin et al. Unmasking Clever Hans predictors and assessing what machines really learn. *Nature Communications*, 2019

9 explanations (LIME), same model, same input…

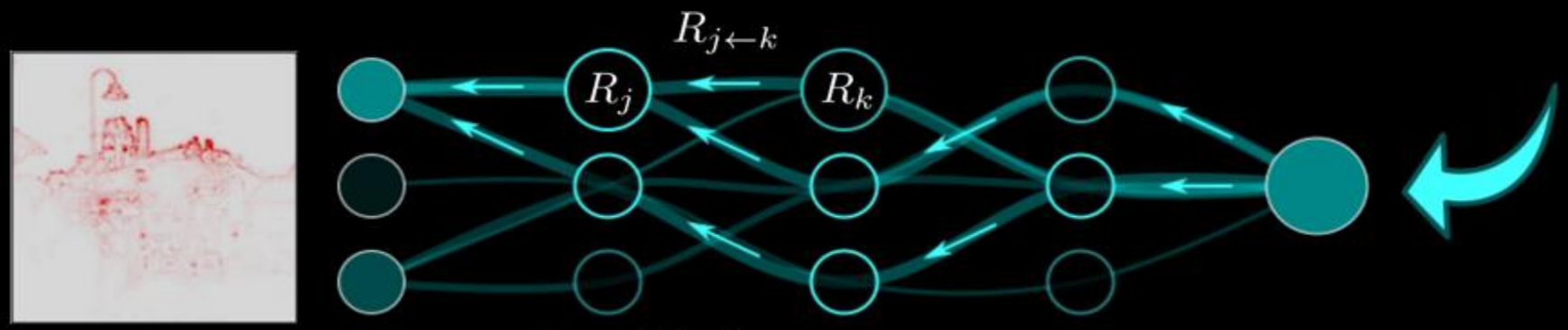
Developing the sensitivity of LIME for better machine learning explanation
Lee et al. 2019

Layer-wise relevance propagation

1. forward pass



2. conservative propagation



$$R_j = \sum_k R_{j\leftarrow k}$$

$$\sum_j R_{j\leftarrow k} = R_k$$

S Bach, A Binder, G Montavon, F Klauschen, KR Müller, W Samek. On Pixel-wise Explanations [...] by Layer-wise Relevance Propagation, PLOS ONE, 10(7):e0130140, 2015

- Making sense of the black box
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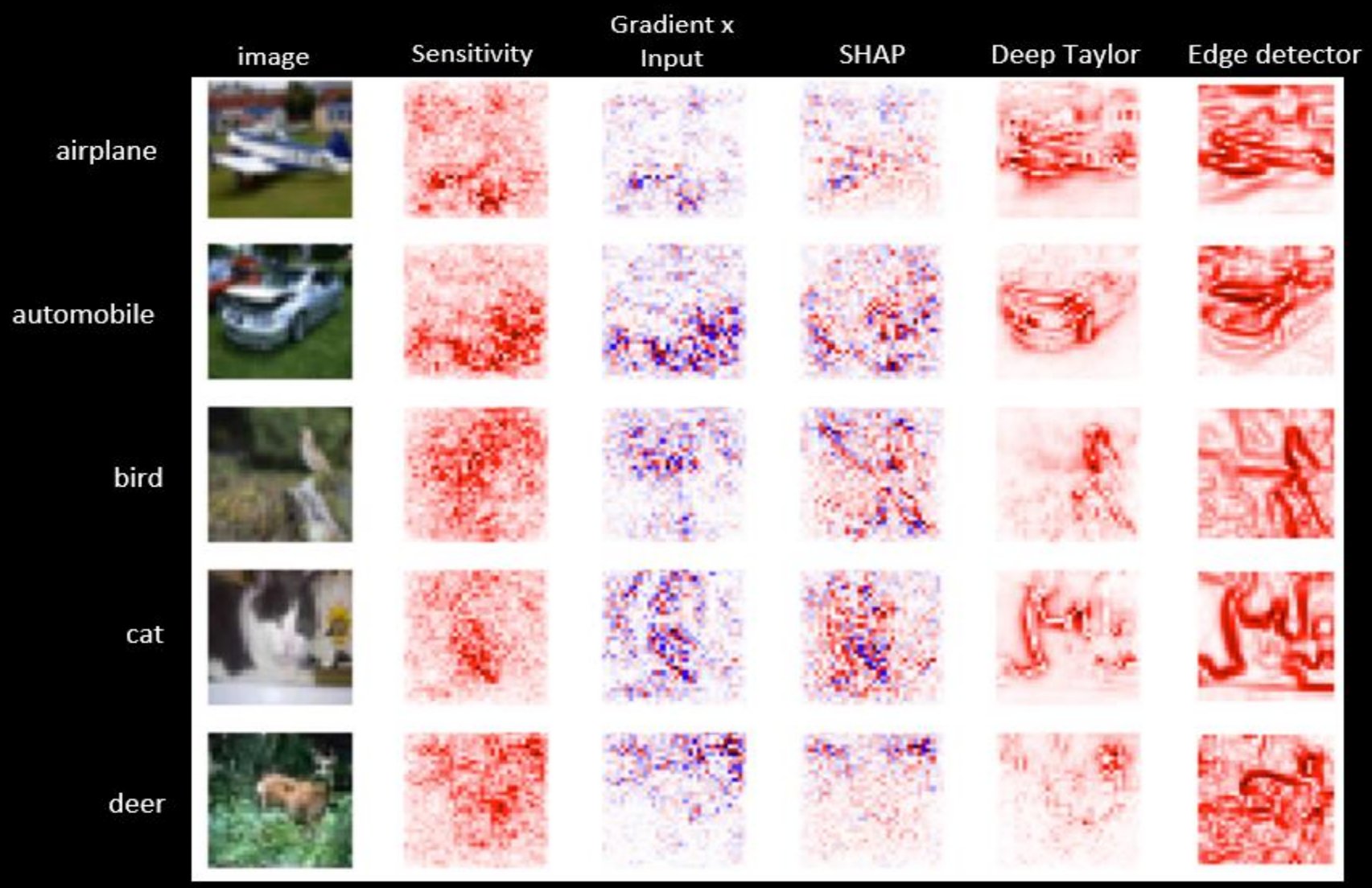


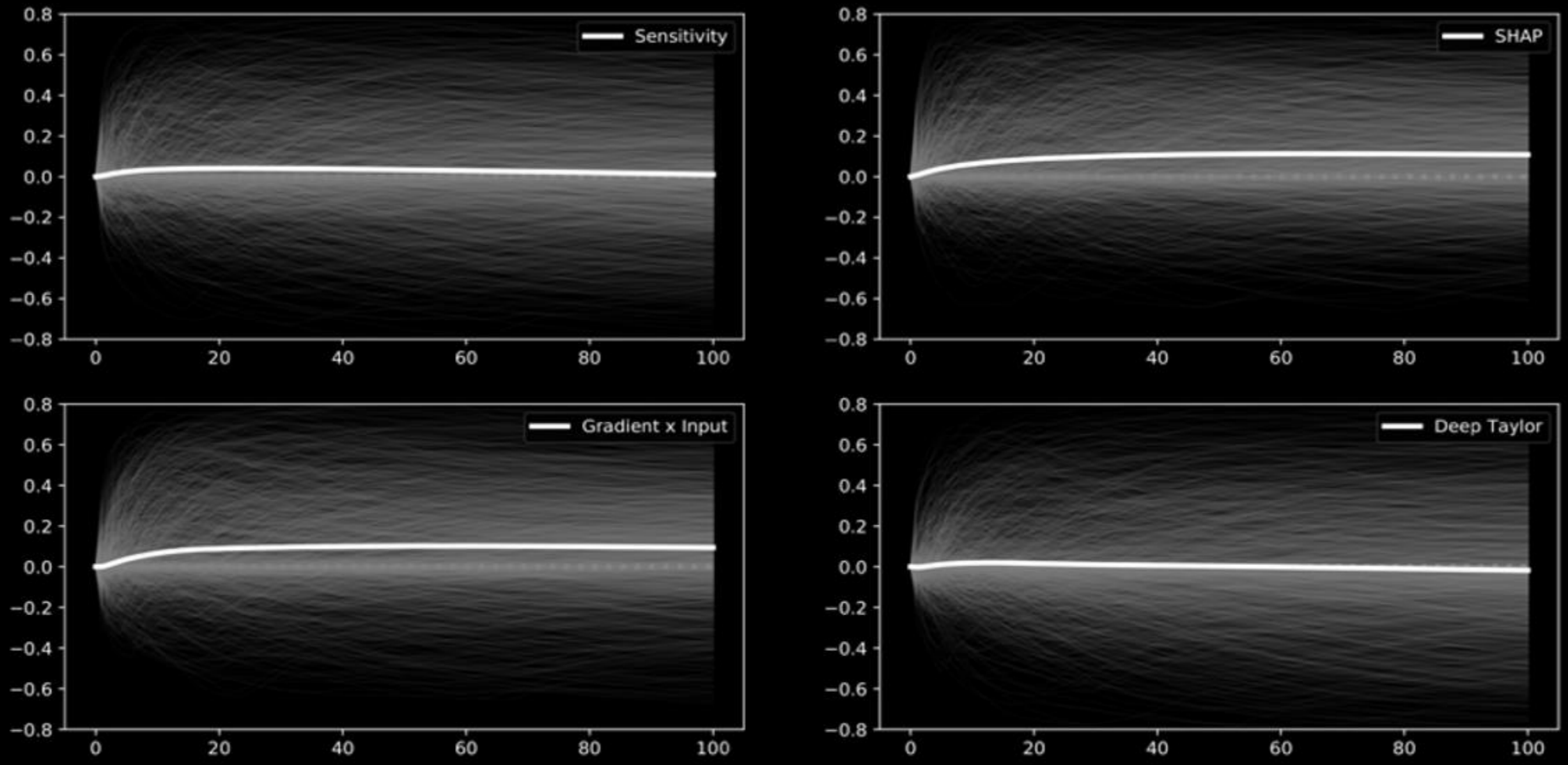
Saliency map explanations for image classification outputs on CIFAR-10 dataset

Area Over Perturbation Curve

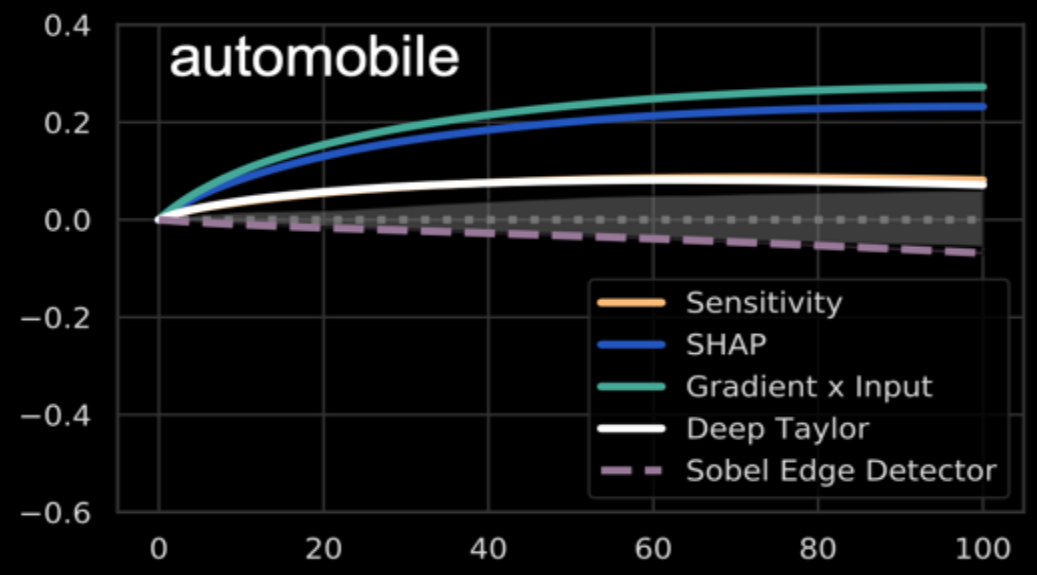
$$\mathrm{AOPC} = \frac{1}{L+1}\left\langle \sum_{k=0}^{L} f(\boldsymbol{x}_{\mathrm{MoRF}}^{(0)}) - f(\boldsymbol{x}_{\mathrm{MoRF}}^{(k)}) \right\rangle_{p(\boldsymbol{x})}$$

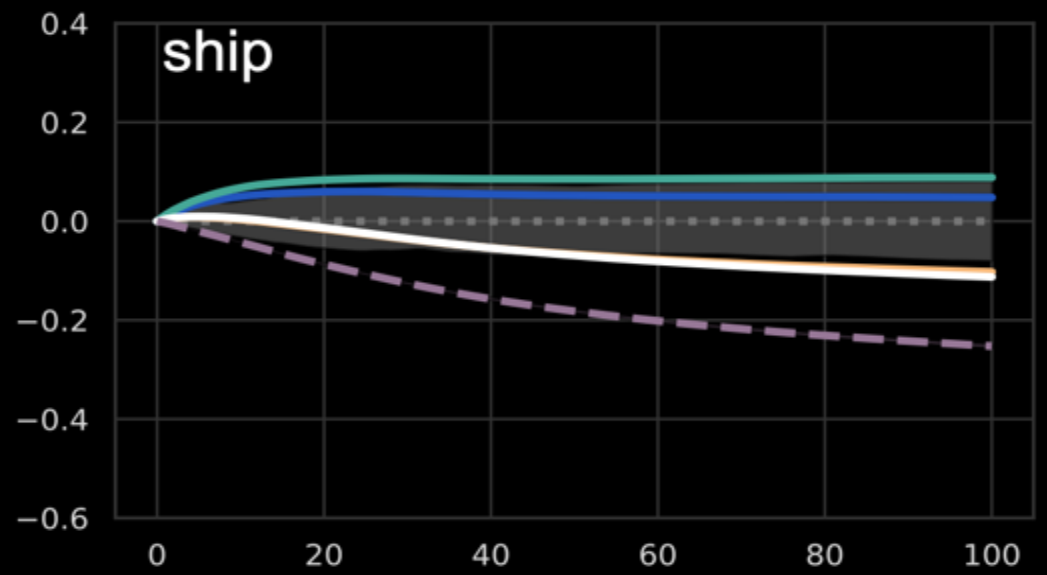
Challenge: How to design a - in this case – a pixel based perturbation method that effectively removes information without introducing (new) spurious structures?

Note: A large AOPC values provides a good explanation of the classifier's decision for a data set.



Sanity Checks for Saliency Metrics
Tomsett et al. 2020 (to appear)





Note: The largest AOPC values best explain the classifier's decision for a data set.

Sanity Checks for Saliency Metrics
Tomsett et al. 2020 (to appear)

- Making sense of the black box
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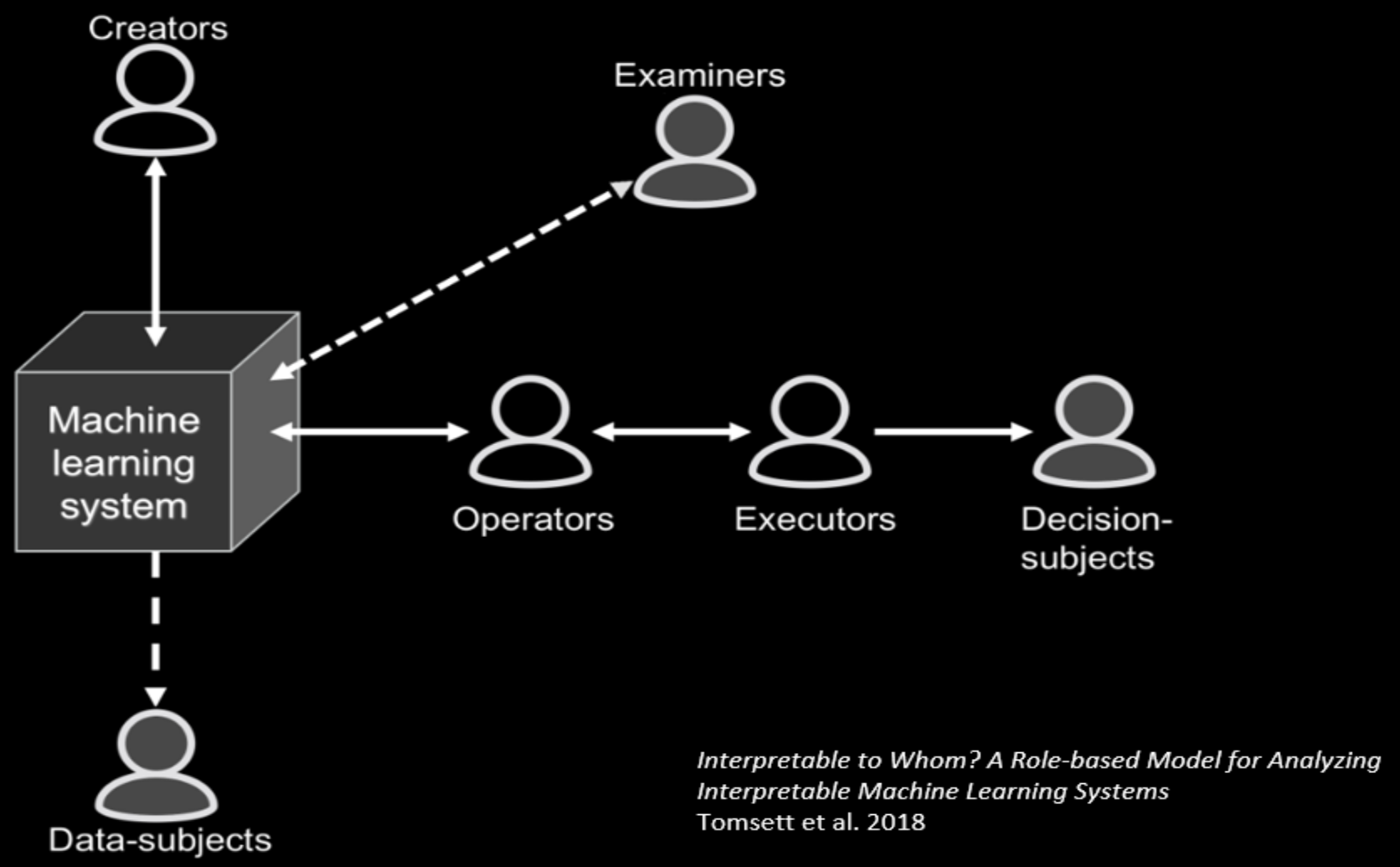
“Interpretable to Whom” framework



Interpretable to Whom? A Role-based Model for Analyzing Interpretable Machine Learning Systems
Tomsett et al. 2018

[Tomsett, R. (2020). Explainable AI (XAI). In *Digital Data Flows Masterclass: Understanding Machine Learning*, Future of Privacy Forum, CPDP 2020 side event]

Tools for Explainability [1]

Triggers	User/Learner's Goal	
	Need to Understand	Need to Accomplish
How do I use it?	Use	Primary task goals
How does it work?	Mechanism, Understandability	Feeling of Satisfaction
What did it just do?	Mechanism, Understandability	Feeling of Sufficiency
What does it achieve?	Function	Usefulness
What will it do next?	Observability	Trust, Reliance
How much effort will this take?	Usability	Primary Task Goals
What do I do if it gets it wrong?	Anticipation	Control
How do I avoid the failure modes?	Surprise	Adaptation, Work-around
What would it have done if x was different?	Mechanism, Understandability	Curiosity, Trust, Reliance
Why didn't it do z?	Anticipation, Surprise	Curiosity, Trust, Reliance

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI, p. 90.

Tools for Explainability [2]

Format (How it is expressed)	Reference (What it is about)
• Visualizations (e.g., Heat maps) • Text (Statements, Narratives or Stories, Answers to queries, Human-machine dialogs) • Formal Expressions (Logical expressions, Matrices) • Conceptual Process Models (Diagrams) • Graphs, Networks • Tables • Abstractions, generalizations • Timelines • Hierarchies (Trees)	• Examples (includes misclassifications, counter-examples, outliers, clear cases, close competitors) • Patterns, Classes, Ontologies • Features, Weights, Probabilities, Ranks, Parameters • Decisions, Strategies, Goals • Algorithms, Computational Processes, Proofs • Incidents, Events (includes self-explanations or stories) • Cause-effect relations

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI, p. 84.

Autoriteit Consument en Markt (10 Dec 2020)

Onderzoek naar toezicht op algoritmische toepassingen

Dit position paper is een vertrekpunt van waaruit de ACM het toezicht op algoritmische toepassingen verder wil ontwikkelen. Het biedt algemene handvatten voor onderzoeken naar overtredingen waarbij algoritmische toepassingen een rol spelen.

De ACM beschrijft in het position paper wat algoritmische toepassing zijn en wanneerzij relevant zijn voor de ACM. Vervolgens licht de ACM toe hoe zij in de praktijk algoritmische toepassingen kan onderzoeken en bespreekt zij enkele uitdagingen waar zij in de praktijk mee te maken kan krijgen.

De ACM draagt met deze publicatie bij aan de nationale en internationale discussie over de toepassing van AI (artificiële intelligentie) door bedrijven en het toezicht daarop door toezichthouders.

Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: *West Virginia Law Review*, *Forthcoming*.

Autoriteit Co 20)

Fairness metrics	Bias preserving?
1. Group fairness, Statistical (demographic) parity[i]	✗
2. Conditional statistical (demographic) parity, Conditional independence[ii]	✗
3. Predictive parity, outcome test[iii]	✓
4. False positive error rate balance[iv]	✓
5. False negative error rate balance,[v] Equal opportunity[vi]	✓
6. Equalized odds[vii]	✓
7. Conditional use accuracy equality[viii]	✓
8. Overall accuracy equality[ix]	✓
9. Treatment equality[x]	✓
10. Test-fairness or calibration[xi]	✓
11. Well-calibration[xii]	✓
12. Balance for positive class[xiii]	✓
13. Balance for negative class[xiv]	✓
14. Causal discrimination[xv] (direct discrimination)	*
15. Fairness through unawareness[xvi]	*
16. Fairness through awareness[xvii]	✗
17. Counterfactual fairness[xviii]	✗
18. No unresolved discrimination[xix] (causal variant of 2)	✗
19. No proxy discrimination[xx]	✗
20. Path based causal reasoning[xxi]	✗

Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: *West Virginia Law Review, Forthcoming*.

AI and data protection risk mitigation and management toolkit

- Gepubliceerd door de Information Commissioner’s Office (ICO).
- De toolkit is bedoeld om risicoprofessionals te helpen de gegevensbeschermingsrisico's die AI-systemen veroorzaken of verergeren, in kaart te brengen en te beperken.
- 13 invalshoeken, waaronder
  - transparantie,
  - statistische nauwkeurigheid,
  - discriminatie en
  - data minimalisatie.

URL: https://ico.org.uk/about-the-ico/ico-and-stakeholder-consultations/ai-and-data-protection-risk-mitigation-and-management-toolkit/.

Number	Risk	How AI can create or exacerbate this risk	Probability (1 = Low, 2 = low/medium, 3 =	Severity (1= low, 2 = low/medium, 3 =	Risk score (probability x severity)	Current status of the risk	Practical steps you could take	Intended action(s)
10,1	Using AI to process personal data without informing individuals impinges on human dignity and a breach of UK GDPR Articles 13, 14 and 22.	AI systems may be opaque, undermining transparency and interpretability of decisions.			0		o Publish privacy information that is concise, transparent, intelligible and uses clear and plain language. o Ensure the privacy information includes meaningful information about the logic involved, as well as the significance and envisaged consequences of the processing for individuals. o Tailor explanations to the intended audience so that they are clear and easy for individuals to understand, considering the level of knowledge that the explanation recipient has about the subject. o Ensure that the privacy information includes the purposes of the processing and the lawful bases (and the legitimate interests for the processing if applicable). o Explain any trade-offs to individuals or any human tasked with reviewing AI output. o Provide privacy information to individuals at the time their personal data is collected (unless an exemption applies). o Ensure individuals are made aware of the privacy information and have an easy way to access it, eg using a combination of appropriate techniques, such as a layered approach, dashboards, just-in-time notices, icons and mobile and smart device functionalities. o Provide details enabling individuals to challenge the outcome if they think it is flawed (eg if some of the input data was incorrect or irrelevant, or additional data wasn't taken into account that the individual thinks is relevant).	
10,2	Failure to provide information to individuals where that information was not obtained from the individual leads to unfair invisible processing, a reduction of human autonomy and a	Datasets containing personal data that are used for training AI purposes may be shared among multiple organisations without the individual's knowledge.					o Carry out due diligence of potential data suppliers to confirm appropriate privacy information has been provided to data subjects. o Carry out a DPIA to determine whether providing privacy information would involve a disproportionate effort when	

... | 5. Lawful Basis | 6. Trade offs | 7. Statistical accuracy | 8. Discrimination | 9. Security & integrity | 10. Transparency | 11. Data minimisation | 12. Individual rights | 13. Human review

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Number	Risk	How AI can create or exacerbate this risk	Probability (1 = Low, 2 = low/medium, 3 =	Severity (1= low, 2 = low/medium, 3 =	Risk score (probability x severity)	Current status of the risk	Practical steps you could take	Intended action(s)
8,1	Failure to review the system's design leads to bias or discriminatory outputs being built into the system	AI systems may lead to unjust discriminatory outcomes if training data is discriminatory.			0		o Ensure that risks are drawn from policy, user research and design, and computer science expertise. o Ensure that the review includes all the relevant checks to the design of the AI systems to ensure that there isn't the potential for discriminatory outputs or decisions to be made. o Ensure that reviews are conducted by appropriately skilled and experienced technical 'experts'. o Document the results of the reviews, including any trade-offs and their technical implications o Ensure that reviews are signed off by senior management. o Document where the AI system design has been adapted / changed following the review where necessary.	
8,2	Failure to use protected characteristics to ensure fairness/accuracy (as defined by the goals of the system) leads to data subjects facing inaccurate or unfair results.	AI systems may use other data points as a proxy for characteristics leading to unlawful discriminatory outcomes that are difficult to detect.			0		o Document considerations and the results of any internal consultations. o Document where possible positive discriminatory characteristics have been included in the system. o Ensure these characteristics are tested thoroughly to ensure they produce the right outputs consistently.	
8,3	No structured testing process in place leads to pre-implementation testing not being undertaken or completed effectively.	There are several different ways of measuring the statistical accuracy of an AI system.			0		o Put in place a policy / documented process that includes details of how the system will be tested prior to implementation. o Ensure the 'test plan' includes all the relevant checks to ensure that there are no discriminatory outputs or decisions (or that these decisions / outputs are at a necessary minimum level) made within AI systems. o Document the results of the testing. o Ensure that the acceptance of the test results is signed off by senior management. o Document where the AI system has been 'retrained' following testing (eg by implementing algorithmic fairness measures / fairness constraints).	

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11,1	A lack of reviews at each stage of the AI lifecycle leads to a risk of inappropriate retention of data and non-compliance with Article 5 (1) (c)	AI systems generally require large amounts of data			0		o Assess whether the features used to train the AI system – and therefore the data – are relevant for the purpose, and make sure in the design stage only that data is processed o Do an assessment to ensure the training data can be modified to reduce the extent to which it can be traced back to specific individuals o Include specific review phases in your development plan to check data is being minimised / not retained when no longer needed. o Do an assessment that includes a justification for the retention of data where applicable o Remove or delete irrelevant data during the system development phase. Maintain a record of what data was removed or deleted.
11,2	Data that is not assessed for its relevance and then separated and deleted leads to excessive data being retained for longer than is necessary and non-compliance with Article 5 (1), (c).	AI systems may find some data points statistically irrelevant.			0		o Map out all the processes in which personal data is used in the different phases of an AI system. o Ensure that the mapping and then subsequent assessment for the potential minimisation of data includes data used in the production of the system and then as part of ongoing research to retrain the system o Index the personal data used in each phase of the AI system lifecycle. o Design a process that detects any duplicated data present in different phases (from production to research) and delete where necessary.

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And finally…

- Aim for surprise minimization!